

B.M.S. COLLEGE OF ENGINEERING, BENGALURU
Autonomous Institute, Affiliated to VTU



**An AAT Report on
on**

“Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data”

**Submitted in fulfilment for the award of degree of
Master of Technology
In
Computer Science and Engineering**

Submitted by

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Under the Guidance of
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B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
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**B.M.S. College of Engineering,
Department of Computer Science and Engineering**



DECLARATION

This is to declare that the AAT work entitled “Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data” carried out by P Yogesh Nayaka (1BM25SCS003-T), is a part of the course Artificial Intelligence (MCS101) of 1st Semester M.Tech (CSE), Department of Computer Science and Engineering, B.M.S. College of Engineering, Bangalore. This AAT work has been carried out under the guidance of Dr. Umadevi V during the academic semester Nov 2025 - Feb 2026. I declare that this work is original and has not been submitted elsewhere for any other degree or award.

Signature of the Candidate

P Yogesh Nayaka (1BM25SCS003-T)

B. M. S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the AAT entitled "Automatic Classification of Sexism in Social Networks: An Empirical Study on Twitter Data" is a bonafide work carried out by P Yogesh Nayaka (1BM25SCS003-T), in partial fulfillment for the award of Master of Technology in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026. This report satisfies the academic requirements prescribed for the said degree.

Guide

Dr. Umadevi V
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1. Introduction:

Social media platforms like Twitter generate a huge amount of text every day. Along with normal conversations, a lot of harmful and biased content also gets shared. One such issue is sexism, which can appear as direct hate or sometimes in very subtle ways. Many sexist tweets do not look offensive at first glance but still promote stereotypes or discrimination.

Manually checking such content is not possible because of the large volume of data. Because of this, machine learning based approaches are useful to automatically detect and classify sexist content. Using Natural Language Processing techniques, text data can be processed and models can be trained to learn patterns in sexist and non-sexist tweets. This project focuses on building such a system using supervised machine learning.

2. Problem Statement:

Sexist content on social media is difficult to detect automatically. This is mainly because sexism is not always explicit. Some tweets contain jokes, sarcasm or indirect statements which makes classification harder. Traditional rule-based systems fail in such cases.

The problem is to design a machine learning model that can correctly classify tweets as sexist or non-sexist using text features. The system should work on short and noisy text like tweets. It should also avoid issues like data leakage and biased evaluation.

3. Objectives:

The main objective of this project is to detect sexism in tweets using machine learning techniques.

To preprocess raw tweet text and convert it into meaningful numerical features.

To train different machine learning models such as Logistic Regression, Support Vector Classifier and Random Forest.

To evaluate and compare the performance of these models using standard evaluation metrics.

4. Dataset Used:

The dataset used in this project is the Sexism Detection in English Texts dataset, it contains English text samples labeled for the presence of sexist content. The dataset has six columns `rewire_id`, `text`, `label_sexist`, `label_category`, `label_vector` and `split`. Table 1 given below will give detailed description on the dataset's attributes.

Column Names	Description	Data Type
<code>rewire_id</code>	Unique identifier for each text instance	String
<code>text</code>	English text sample	String
<code>label_sexist</code>	Binary label indicating sexist or non-sexist content.	Categorical
<code>label_category</code>	Type of sexism	Categorical
<code>label_vector</code>	Fine-grained sexism annotation	Categorical
<code>split</code>	Dataset split used for training or evaluation	Categorical

Table 1: Dataset Description

Each row has a unique identifier `rewire_id` which is a text segment. The `text` column has the tweets which are processed and used for training classic machine learning models. The `label_sexist` is the target column which has two classes: sexist and not sexist. With these columns the dataset also has `label_category` and `label_vector`, which describe the type of sexism. The dataset has a total of 20,000 rows, which are divided into training and testing sets. The training set has 16,000 rows in which 3,884 rows are labeled sexist and 12,116 rows labels are non-sexist. The testing set has 4,000 rows in which 970 rows are labeled sexist and remaining 3,030 are labeled non-sexist. Table 2 below shows how train and test are split.

Class Label	Train	Test
Sexist	3,884	970
Not-Sexist	12,116	3,030

Table 2: Dataset Split

5. Tools Used for Implementation:

5.1 Programing Language:

- **Python**

5.2 Python Libraries:

- **Pandas**
- **Matplotlib**
- **Seaborn**
- **SciKit-Learn**

5.3 Text Cleaning Libraries:

- **Regular Expression(re)**
- **String**
- **Emoji**
- **NLTK**

5.4 Machine Learning Models:

- **Logistic Regression**
- **Support Vector Classifier(SVC)**
- **Random Forest Classifier**

6. Github Link: <https://github.com/nayakayogesh/Twitter-Text-Classification>

7. Methodology:

This project follows the standard machine learning pipeline. First, loading and exploring the data. Pandas is used to understand the data and see class distribution. Then raw text was cleaned by lowercasing, removing punctuation, URLs, emojis, stopwords and stemming to reduce word to root form.

The data was split into training and testing split, the split made was 80/20. To prevent data leakage a pipeline is used with TF-IDF and machine learning models. The models chosen were Logistic Regression, Support Vector Classifier and Random Forest.

These models were trained on TF-IDF features in the pipeline. After training these models were evaluated using accuracy, precision, recall, F1-score and ROC-AUC. Figure 1 below shows the Proposed Methodology for Text Classification.

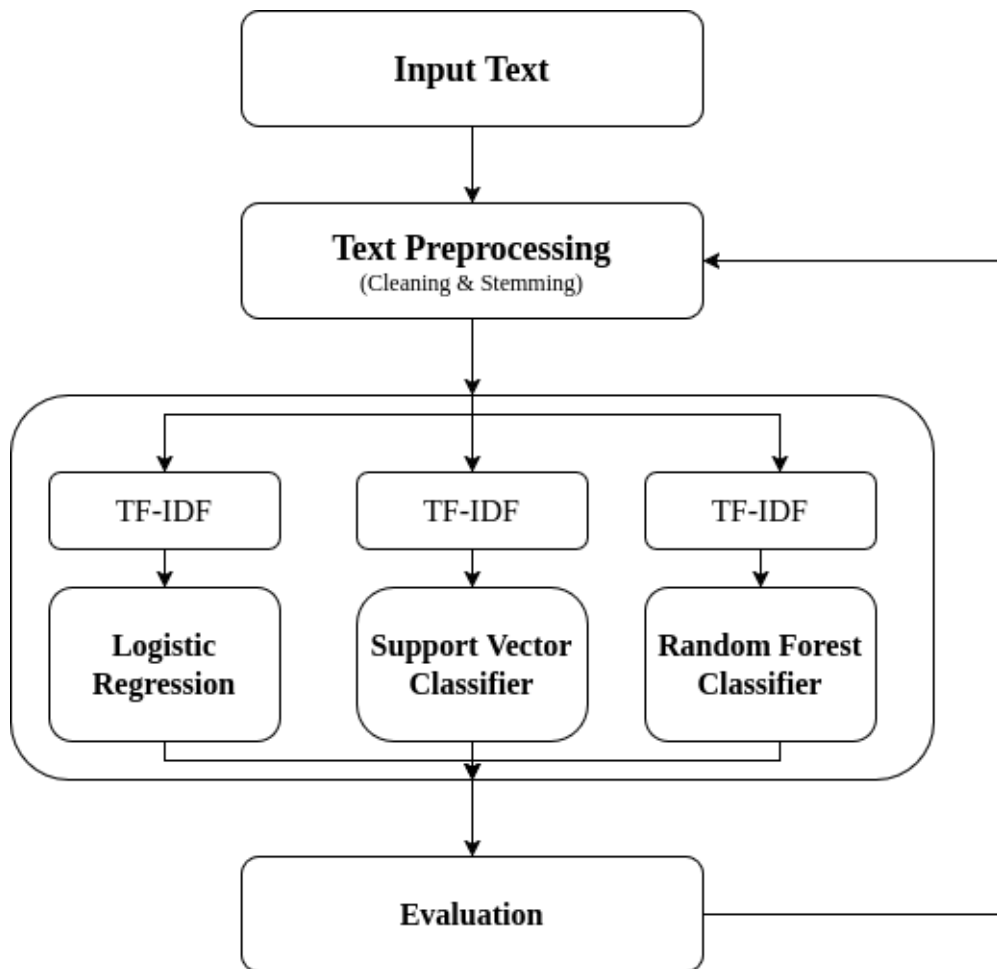


Figure 1: Methodology Flowchart for Text Classification

8. Results Obtained:

The trained models were evaluated on the test dataset using different metrics like accuracy, precision, recall, F1-score and ROC-AUC.

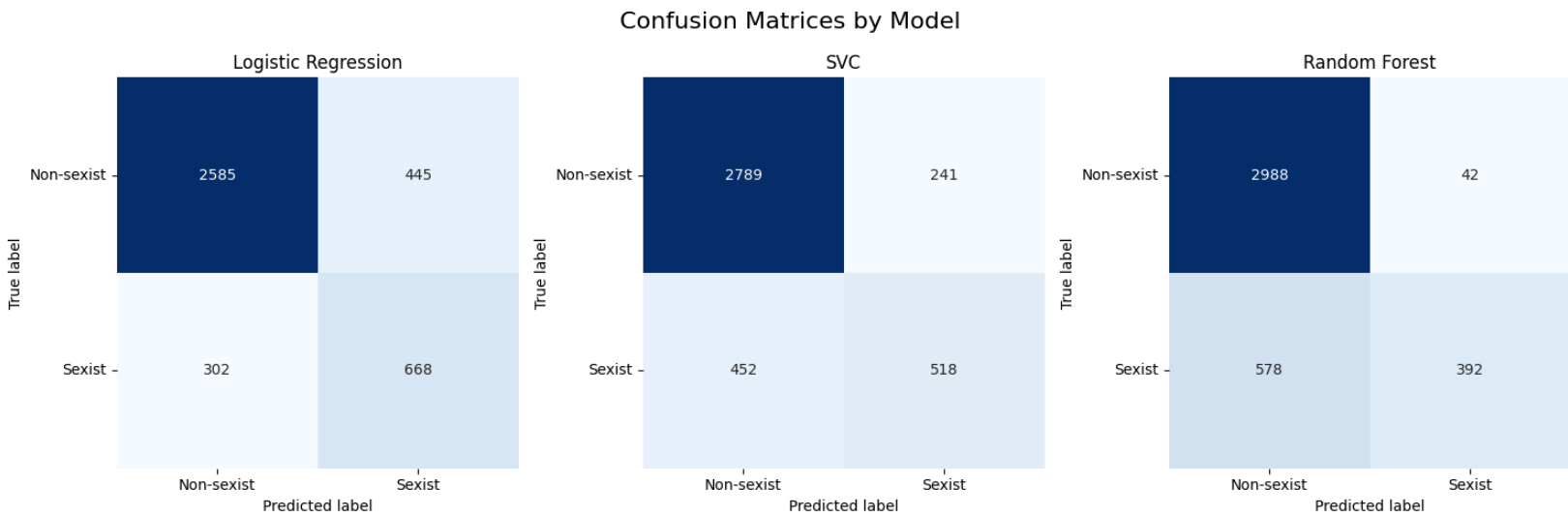


Figure 2: Confusion Matrix of each models

Figure 2 shows the confusion matrix of each model used. The confusion matrix shows that all three models have the same True Negative, but Logistic Regression has better True Positive and Random Forest is worse than the other models. Analysing further random forest, False Negative is high which is not ideal while detecting sexist content.

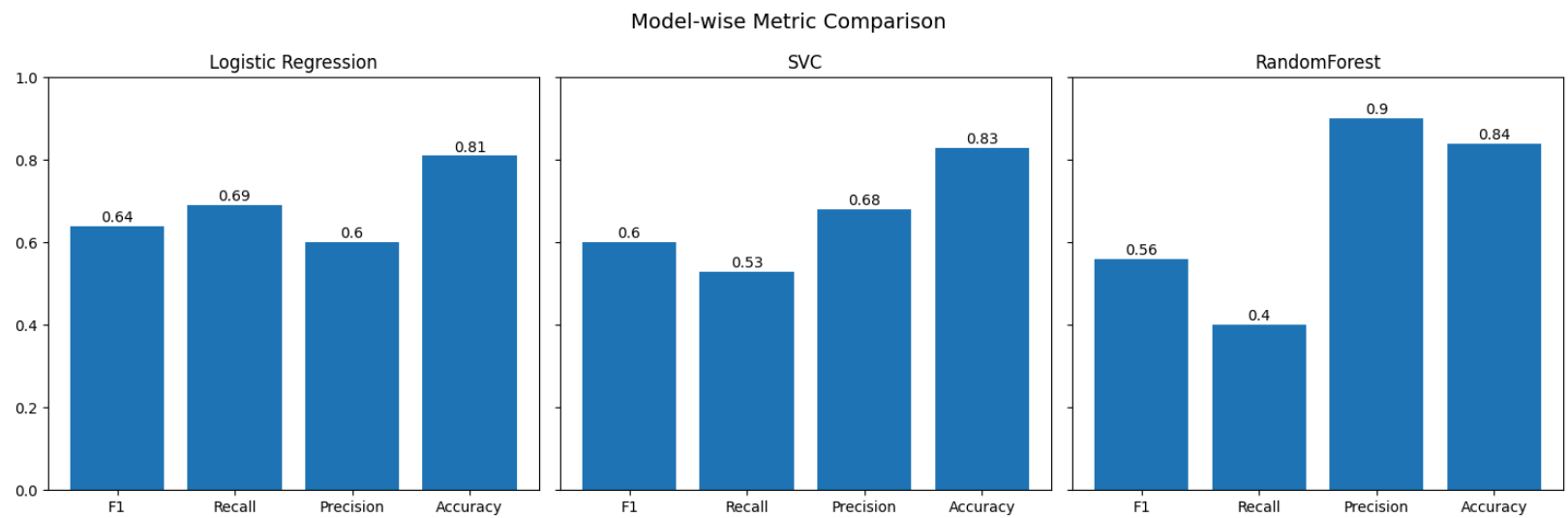


Figure 3: Metric Comparison of each model

Figure 3 shows metrics of each model used. Even though the accuracy of Random Forest is high compared to Logistic Regression and Support Vector Classifier, due to the highly imbalanced dataset and context of the dataset recall is used to catch all the sexist text.

Logistic Regression has recall of 0.69, Support Vector Classifier has recall of 0.53 and Random Forest has recall of 0.4. This suggests that Logistic Regression performs moderately on the given dataset. The lower values of Support Vector Classifier and Random Forest shows that the models miss large amounts of instances. Table 4 shows comparison of model metrics with those reported in the reference paper.

Model	Paper Result				Implemented Result			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
LR	0.71	0.62	0.64	0.62	0.81	0.6	0.69	0.64
SVC	0.71	0.61	0.63	0.61	0.83	0.68	0.53	0.60
RFC	0.72	0.67	0.57	0.6	0.84	0.9	0.4	0.56

Table 4: Comparison between paper and implementation

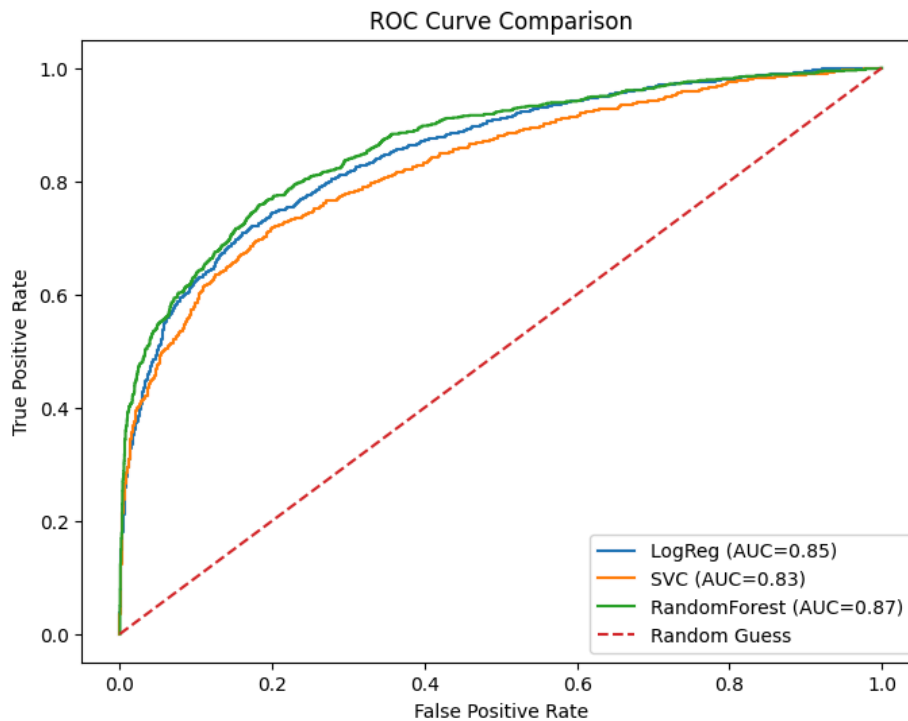


Figure 4: ROC-AUC of the models

Figure 4 shows ROC-AUC of the models. Random Forest, Logistic Regression and Support Vector Classifier has ROC-AUC of 0.87, 0.85 and 0.83 respectively. Although Random Forest has the highest ROC-AUC, its low recall was low for sexist content. Even though Logistic Regression was slightly low ROC-AUC, it achieved better recall and F1-score for sexist class.

9. Learning Outcome:

Through this project, a practical understanding of Natural Language Processing and text classification was developed. Highlights include text preprocessing, feature extraction using TF-IDF and model selection for respective tasks. The project also provided hands-on experience with evaluating machine learning models using different metrics, not just accuracy.

This project helped understanding of applying machine learning techniques to real-world social issues like sexism detection in text data.

10. Paper Reference:

- Rodríguez-Sánchez, F., Carrillo-de-Albornoz, J. and Plaza, L., 2020. Automatic classification of sexism in social networks: An empirical study on twitter data. IEEE Access, 8, pp.219563-219576.

11. Screenshot of Similarity & AI Generated report:

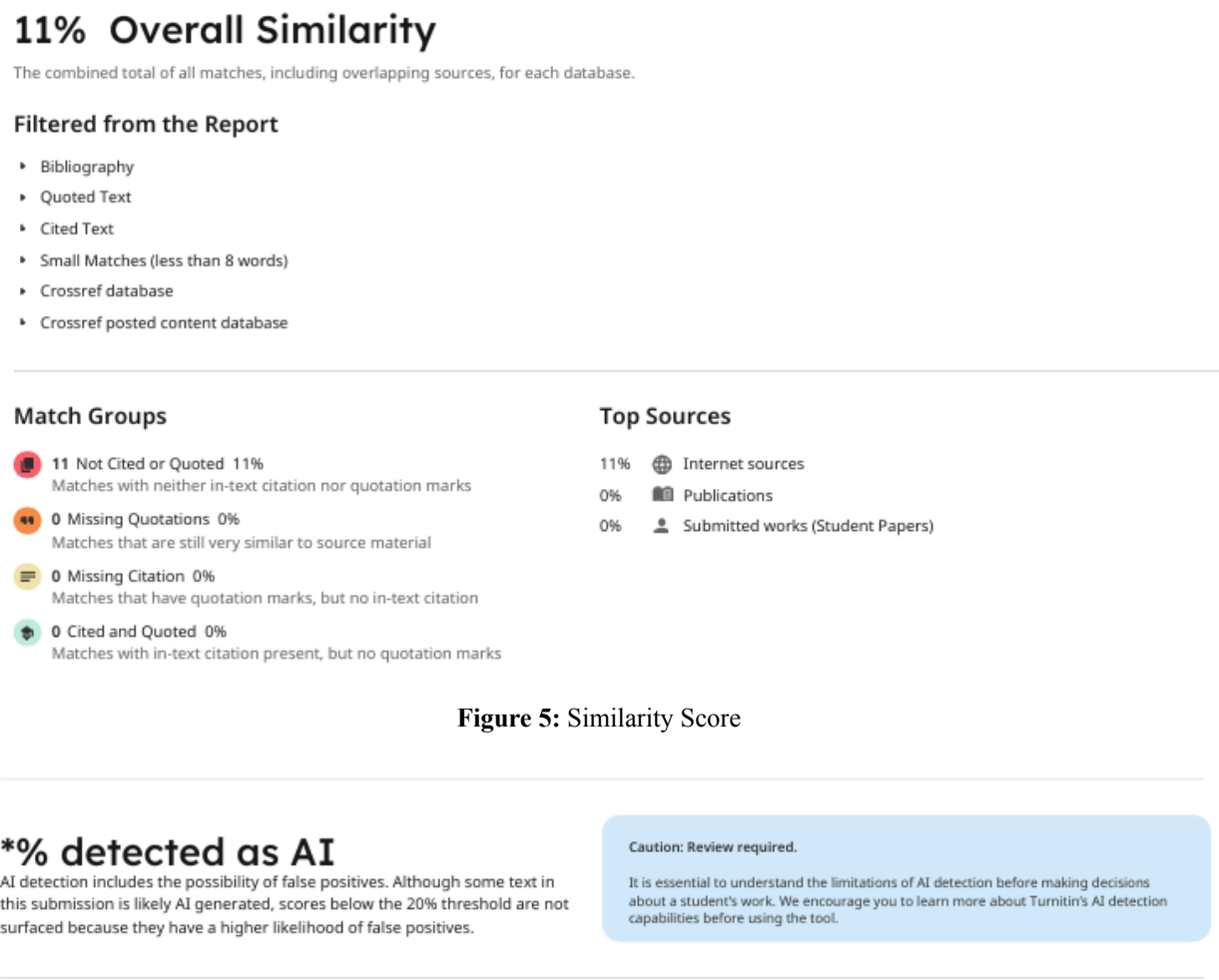


Figure 6: AI Report